ANALYSIS REPORT

CLINICAL TRIAL TREATMENT EFFECTIVENESS AND SAFETY ANALYSIS

Executive Pharma Analytics Report

Tools Used: Python, PostgreSQL, Power BI

Dataset: 52,321 chemotherapy clinical trial patient records

1. Executive Summary

This project evaluates chemotherapy treatment effectiveness and patient safety using 52,321 patient records. The analysis combines Python-based data preparation, PostgreSQL query outputs, statistical analysis, and Power BI dashboarding. The overall positive response rate is 60.06%, average survival is 62.53 months, severe side-effect rate is 20.10%, neutropenia rate is 39.93%, and metastasis rate is 29.92%.

The updated project strengthens the analysis with additional SQL outputs covering tumor response distribution, complete response rate, partial response matrix, neutropenia rate, cancer-type distribution, metastasis by tumor stage, and high-risk patient rate. These outputs provide stronger evidence for treatment effectiveness, patient safety, and segmentation analysis.

2. Business Objective

The objective is to identify treatment effectiveness, adverse event patterns, and high-risk patient groups across chemotherapy regimens. The project supports clinical trial monitoring by translating raw patient-level data into KPIs, SQL evidence, and an interactive Power BI dashboard.

3. Dataset Profile

Attribute	Value	Interpretation
Total Records	52,321	Large patient-level dataset suitable for segmentation and dashboard analysis.
Total Source Columns	17	Covers demographics, disease profile, treatment exposure, safety, and outcomes.
Average Age	57.05 years	Represents a mature oncology population.
Average BMI	26.75	Supports patient health profiling.

Average Dosage	325.35 mg/m2	Useful for dosage and exposure-related analysis.
Average Cycles Completed	4.00	Supports treatment completion and exposure review.
Average Survival	62.53 months	Core effectiveness outcome for clinical interpretation.

4. Data Preparation and Feature Engineering

Python was used to clean the data, standardize column names, handle missing values, create derived analytical columns, and prepare the dataset for PostgreSQL and Power BI. Missing genetic mutation values were categorized as unknown, and missing chemotherapy regimen values were categorized as not assigned.

- Created age-group segments for demographic analysis.
- Created positive response flag using Complete and Partial tumor responses.
- Created severe side-effect flag using nausea severity greater than or equal to 4.
- Created high-risk patient logic using tumor stage, metastasis, and adverse event indicators.
- Loaded the cleaned dataset into PostgreSQL for SQL-based analysis.

5. KPI Summary

Total Patients

52,321

Average Age

57.05

Overall Response Rate

60.06%

Complete Response Rate

20.07%

Partial Response Rate

39.99%

Average Survival

62.53 mo.

Severe Side Effect Rate

20.10%

Neutropenia Rate

39.93%

6. Python and Statistical Analysis

Python was used for data cleaning, exploratory data analysis, KPI calculation, and visualization. The notebook also includes statistical testing using chi-square analysis for treatment regimen versus tumor response and ANOVA for survival differences across chemotherapy regimens. These tests add analytical depth beyond dashboard reporting.

7. SQL Analysis Outputs

PostgreSQL was used to validate and summarize clinical trial metrics. The updated project includes multiple SQL output screenshots, making the database analysis more complete and evidence-driven.

7.1 Clinical Trials Table Preview

	patient_id text	age bigint	sex text	bmi double precision	smoking_status text	cancer_type text	genetic_mutation text	tumor_stage text	tumor doubl
1	P00001	68	Male	31.5	Former	Breast	BRCA1	II	
2	P00002	81	Fema...	25.8	Former	Lung	KRAS	I	
3	P00003	58	Male	22.3	Former	Lymphoma	BRCA1	II	
4	P00004	44	Male	33.6	Never	Lymphoma	EGFR	IV	
5	P00005	72	Male	23.7	Never	Breast	TP53	III	
6	P00006	37	Male	35	Never	Breast	KRAS	III	
7	P00007	50	Male	31.2	Never	Lung	Unknown	IV	
8	P00008	68	Fema...	29.8	Current	Colon	TP53	II	
9	P00009	48	Male	18.7	Never	Lymphoma	TP53	II	
10	P00010	52	Fema...	25.9	Current	Leukemia	KRAS	II	
11	P00011	40	Male	26.8	Never	Colon	TP53	III	
12	P00012	40	Fema...	24.7	Former	Lymphoma	TP53	IV	
13	P00013	53	Male	21	Never	Lung	TP53	III	
14	P00014	82	Fema...	24.6	Never	Lymphoma	BRCA1	I	
15	P00015	65	Male	19.7	Never	Colon	EGFR	II	
16	P00016	69	Male	23.3	Never	Leukemia	EGFR	I	
17	P00017	53	Male	19.9	Never	Colon	EGFR	II	
18	P00018	32	Male	30.9	Never	Colon	TP53	III	
19	P00019	51	Fema...	29.6	Former	Breast	TP53	I	

Figure 1: Cleaned clinical trial table loaded into PostgreSQL.

7.2 Tumor Response Distribution

	tumor_response text	total_patients bigint	percentage numeric
1	Partial	20925	39.99
2	Stable	15688	29.98
3	Complete	10501	20.07
4	Progressive	5207	9.95

Figure 2: Tumor response distribution from SQL output.

Partial response is the largest outcome group with 20,925 patients, representing 39.99% of the population. Complete response accounts for 20.07%, stable disease for 29.98%, and progressive disease for 9.95%.

7.3 Complete Response Rate by Chemotherapy Regimen

	chemotherapy_regimen text	total_patients bigint	complete_responses bigint	complete_response_rate numeric
1	Not Assigned	2556	539	21.09
2	Gemcitabine	10375	2142	20.65
3	ABVD	13052	2642	20.24
4	FOLFOX	15699	3113	19.83
5	CHOP	10639	2065	19.41

Figure 3: Complete response rate by chemotherapy regimen.

The not assigned group shows the highest complete response rate at 21.09%, followed by Gemcitabine at 20.65% and ABVD at 20.24%. CHOP records the lowest complete response rate among the listed regimens at 19.41%.

7.4 Treatment Outcome Matrix

	chemotherapy_regimen text	complete_response bigint	partial_response bigint	stable_disease bigint	progressive_disease bigint
1	ABVD	2642	5241	3881	1288
2	CHOP	2065	4290	3231	1053
3	FOLFOX	3113	6251	4783	1552
4	Gemcitabine	2142	4146	3055	1032
5	Not Assigned	539	997	738	282

Figure 4: Treatment outcome matrix by chemotherapy regimen.

The treatment outcome matrix compares complete, partial, stable, and progressive disease counts across regimens. FOLFOX has the largest patient volume and therefore the highest absolute counts across most outcome categories.

7.5 Neutropenia Rate by Chemotherapy Regimen

	chemotherapy_regimen text	total_patients bigint	neutropenia_cases bigint	neutropenia_rate numeric
1	Gemcitabine	10375	4250	40.96
2	CHOP	10639	4258	40.02
3	Not Assigned	2556	1014	39.67
4	FOLFOX	15699	6208	39.54
5	ABVD	13052	5160	39.53

Figure 5: Neutropenia rate by chemotherapy regimen.
Gemcitabine has the highest neutropenia rate at 40.96%, followed by CHOP at 40.02%. Neutropenia remains an important safety KPI because nearly 40% of patients experience this adverse event.

7.6 Patient Distribution by Cancer Type

	cancer_type text	total_patients bigint	patient_percentage numeric
1	Colon	10522	20.11
2	Leukemia	10478	20.03
3	Lymphoma	10469	20.01
4	Breast	10458	19.99
5	Lung	10394	19.87

Figure 6: Patient distribution by cancer type.
Cancer types are evenly distributed, with Colon cancer at 20.11%, Leukemia at 20.03%, Lymphoma at 20.01%, Breast cancer at 19.99%, and Lung cancer at 19.87%. This supports balanced high-level comparison across cancer categories.

7.7 Metastasis Rate by Tumor Stage

	tumor_stage text	total_patients bigint	metastasis_cases bigint	metastasis_rate numeric
1	II	18203	5493	30.18
2	IV	7852	2370	30.18
3	I	13045	3892	29.84
4	III	13221	3902	29.51

Figure 7: Metastasis rate by tumor stage.
Tumor Stage II and Stage IV show the highest metastasis rates at 30.18%, followed by Stage I at 29.84% and Stage III at 29.51%. Metastasis status is a critical risk dimension for clinical monitoring.

7.8 High-Risk Patient Rate by Age Group

	age_group text	total_patients bigint	high_risk_patients bigint	high_risk_patient_rate numeric
1	30-45	14255	9606	67.39
2	61-75	14025	9362	66.75
3	46-60	14397	9564	66.43
4	75+	8744	5761	65.89
5	<30	900	582	64.67

Figure 8: High-risk patient rate by age group.

The 30-45 age group shows the highest high-risk patient rate at 67.39%, followed by the 61-75 age group at 66.75%. This reinforces the need to review patient risk across age bands rather than assuming risk is limited only to older patients.

7.9 Average Survival by Tumor Stage

	tumor_stage text	total_patients bigint	avg_survival_months numeric
1	III	13221	62.66
2	I	13045	62.66
3	II	18203	62.55
4	IV	7852	62.04

Figure 9: Average survival months by tumor stage.

7.10 Severe Side Effect Rate by Age Group

	age_group text	total_patients bigint	avg_nausea_severity numeric	severe_side_effect_rate numeric
1	75+	8744	2.40	20.45
2	46-60	14397	2.40	20.25
3	30-45	14255	2.41	20.18
4	61-75	14025	2.39	19.66
5	<30	900	2.37	19.56

Figure 10: Severe side-effect rate by age group.

8. Power BI Dashboard

Figure 11: One-page Power BI dashboard for treatment effectiveness and safety monitoring.

The Power BI dashboard presents a one-page executive view of the clinical trial analysis. It includes KPI cards, cancer type and regimen slicers, severe side-effect analysis, neutropenia cases, patient distribution across tumor stages, tumor response distribution, and chemotherapy response performance.

9. Key Business Insights

- The overall positive response rate is 60.06%, combining complete and partial responses.
- Partial response is the dominant outcome category at 39.99%, while complete response accounts for 20.07%.
- Progressive disease represents 9.95% of patients and should be monitored as a treatment failure indicator.
- Gemcitabine shows the highest neutropenia rate at 40.96%, making it important for safety review.
- Patient distribution across cancer types is well balanced, supporting meaningful cross-segment comparison.
- Stage IV has the lowest average survival among tumor stages, supporting focused monitoring of advanced-stage patients.
- High-risk patient rates are above 64% across all age groups, indicating that risk management is relevant across the full trial population.

10. Recommendations

- Monitor treatment response and survival together to avoid relying on a single effectiveness metric.
- Use neutropenia and severe side-effect rates as core safety indicators in ongoing trial review.
- Segment patients by cancer type, tumor stage, age group, and regimen to identify high-risk cohorts.
- Review regimens with stronger response metrics alongside their adverse event rates before making business or clinical interpretations.
- Extend the analysis with survival curves, dosage-response modeling, and protocol-level variables if richer clinical trial data becomes available.

11. Limitations

This project is suitable for analytical demonstration and portfolio presentation. Clinical recommendations should not be made without validated protocol information, randomization details, eligibility criteria, adverse event grading standards, statistical significance interpretation, and medical review.

12. Conclusion

This project demonstrates a complete pharma analytics workflow using Python, PostgreSQL, and Power BI. The analysis provides a strong view of treatment effectiveness, tumor response, survival outcomes, adverse events, metastasis patterns, and high-risk patient segmentation. The final dashboard and SQL outputs create a professional, evidence-backed project suitable for pharmaceutical analytics, healthcare analytics, and business intelligence portfolios.